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SIMATS ENGINEERING
ASSESSMENT TOOL 2

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192524214

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

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1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile

devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for

redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		contribution (if team-based)			

	A	B	C	D	E	F	G	H	I	
Q ₁	1.5	2	2	1	0.5	0.5	0.5	0.5	0.5	9
Q ₂	1	1.5	1.5	0.5	1	0.5	0.5	0.5	0.5	7.5
Q ₃	1.5	2	2	0.5	1	1	0.5	0.5	0.5	7.5
Q ₄	1	1	1	1	1	1	0.5	0.5	0.5	9.5
Q ₅	1.5	2	2	1	0.5	1				

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1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

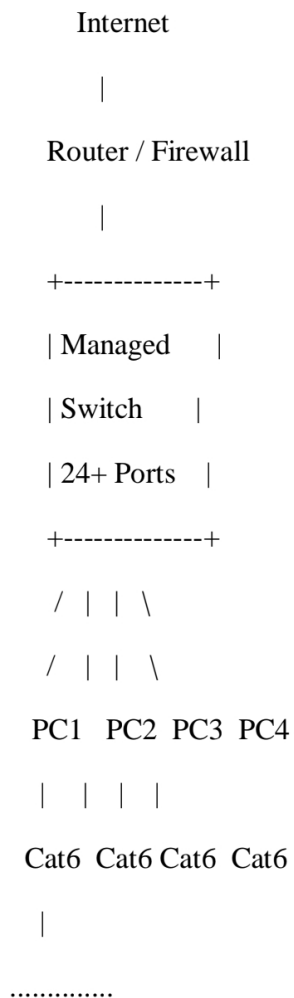
ANS: . Aim

To design and explain a star topology network suitable for a 20-person startup, using a central managed switch to connect computers, printers, VoIP phones, and wireless devices.

2. Network Components

Component	Quantity	Purpose
Managed Switch	1	Central connection point for all devices
Cat6 UTP Cable	20+	Connects devices to the switch
Router/Firewall	1	Internet access, DHCP and security
Wi-Fi Access Point	1	Provides wireless connectivity
Workstations	20	Employee computers
Printer	As required	Network printing
VoIP Phones	As required	Voice communication

3. Topology Diagram



|
PC20

Wi-Fi Access Point

|
Laptops / Mobiles

4. Working

- The managed switch acts as the central device.
- Each workstation is connected directly to the switch using Cat6 UTP cable.
- The switch forwards data between connected devices.
- The router/firewall connects the internal network to the Internet.
- DHCP can automatically assign IP addresses to network devices.
- A Wi-Fi access point provides wireless connectivity.
- If one cable or workstation fails, the remaining devices continue to operate.

5. Advantages

1. Easy to manage because all devices connect to one central switch.
2. Easy troubleshooting because faults can be isolated to individual connections.
3. Fault isolation—failure of one cable affects only one device.
4. High performance with Gigabit Ethernet over Cat6.
5. Easy expansion because unused switch ports can be used for additional devices.
6. Better security and control using a managed switch.
7. Suitable for offices with computers, printers, VoIP phones, and Wi-Fi devices.

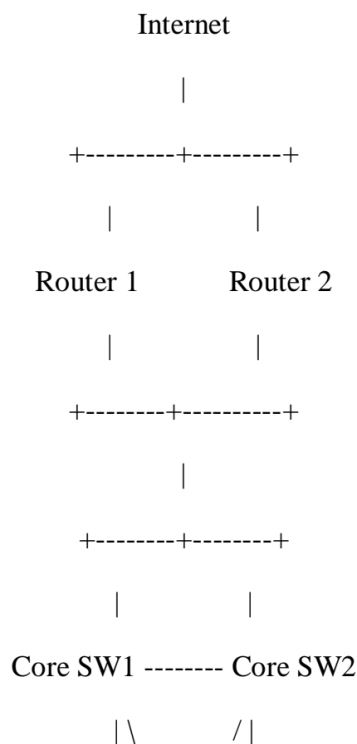
. Conclusion

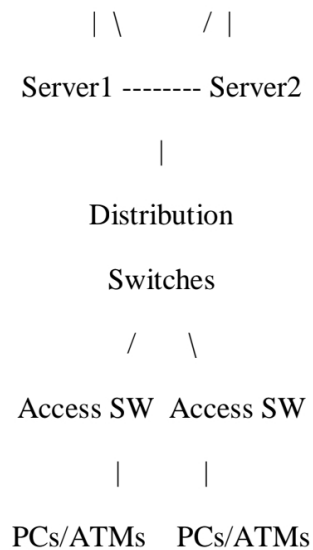
A star topology is an effective choice for a 20-person startup because it provides centralized management, easy troubleshooting, good performance, scalability, and fault isolation. A 24-port or larger managed switch provides enough connections for current devices while leaving ports available for future expansion.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

ANS: AIM: To design a highly reliable network for a bank that provides near-zero downtime, redundancy, high availability, and secure communication.

Network Design:





Working

- Core routers, switches, and critical servers are interconnected using redundant links.
- A partial mesh is normally preferred because it provides redundancy without the excessive cabling of a full mesh.
- Two enterprise routers provide redundant Internet connections.
- OSPF/BGP can automatically select alternative routes when a link fails.
- Fiber-optic cables provide high bandwidth and low latency.
- Redundant power supplies, UPS systems, and RAID storage improve overall availability.

Advantages

1. Very high reliability.
2. Multiple alternative communication paths.
3. Failure of one link does not stop network operation.
4. Supports automatic traffic rerouting.
5. Suitable for critical banking applications.
6. Provides high bandwidth and low latency.

Limitation

The main disadvantages are high installation cost, complex configuration, and increased cabling requirements.

Conclusion

A partial mesh topology is highly suitable for banking networks because it provides redundancy and automatic failover while avoiding the excessive cost of a complete full-mesh network.

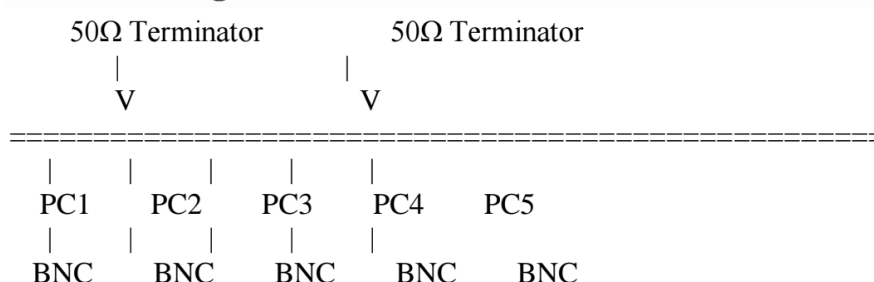
3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

ANS:

Aim

To design a low-cost network for a small classroom computer laboratory where hardware and installation costs must be minimized.

Network Design:



Working:

- All computers share a **single backbone cable**.
- In a traditional bus network, coaxial cable such as **RG-58** can be used.
- BNC connectors and T-connectors connect computers to the backbone.

- Terminators are placed at both ends of the cable to prevent signal reflection.
- Data transmitted by one computer travels along the shared bus.
- All connected computers can receive the transmitted signal, but only the intended computer processes it.

Advantages

1. Low installation cost.
2. Requires less cable.
3. No dedicated central switch is required in the traditional design.
4. Simple structure.
5. Suitable for small networks with light traffic.

Limitations

1. A backbone cable failure can affect the entire network.
2. Network performance decreases as more devices are added.
3. Collision problems can occur.
4. Troubleshooting can be difficult.
5. Limited scalability.

Conclusion

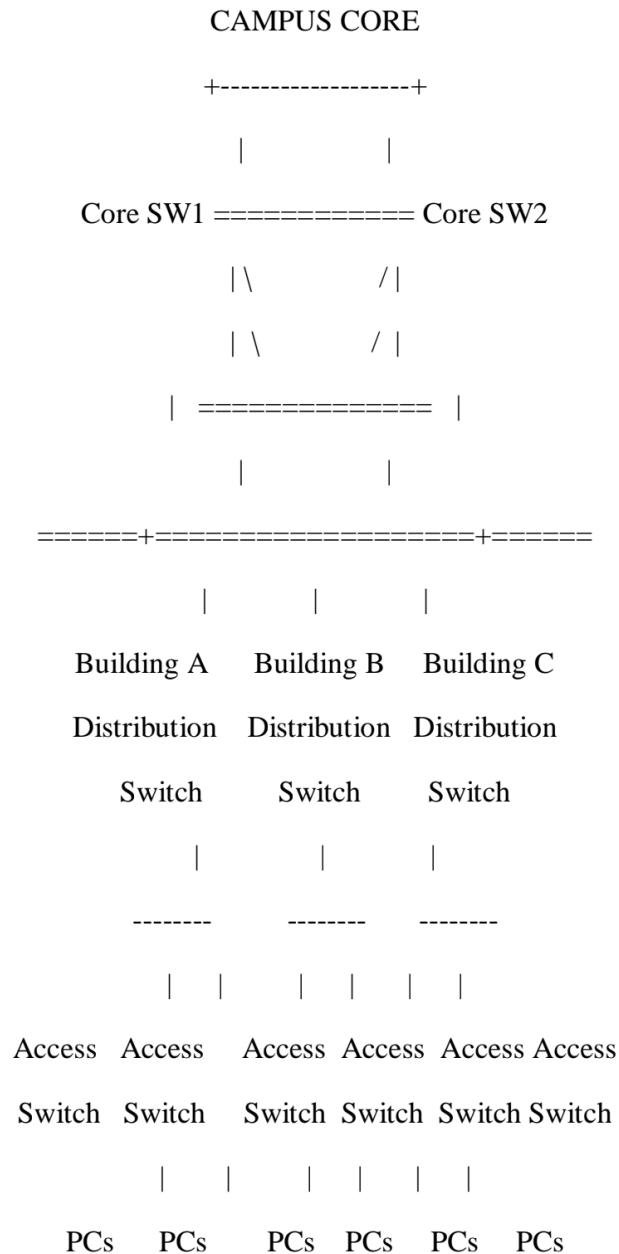
Bus topology can be considered for a **small, low-budget classroom environment** where traffic is light. However, modern Ethernet networks generally use switches rather than traditional bus cabling because switched Ethernet is more reliable and scalable.

4. **A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.**

Ans:

Aim: To design a scalable, reliable, and high-performance network for a large university campus containing multiple buildings and thousands of users.

Network Design:



Working

- The **campus core** uses redundant fiber links between core switches.
- Each building has distribution switches connected to the core using **dual fiber uplinks**.

- Floor-level access switches connect computers and other devices.
- Cat6 cables can be used between access switches and end devices.
- Wireless access points connect to switches using **PoE**.
- If one backbone link fails, traffic can use another available path.
- New buildings can be added without redesigning the entire network.

Advantages

1. Highly scalable.
2. Supports thousands of users.
3. Provides backbone redundancy.
4. Easy building and department expansion.
5. Centralized network management.
6. High-speed fiber connectivity.
7. Supports wired and wireless devices.

Limitations:

- Higher installation cost.
- Requires skilled network administrators.
- Core network design and maintenance are more complex.

Conclusion

A hybrid topology is ideal for a university because it combines the redundancy of mesh/ring, manageability of star, and scalability of hierarchical networks.

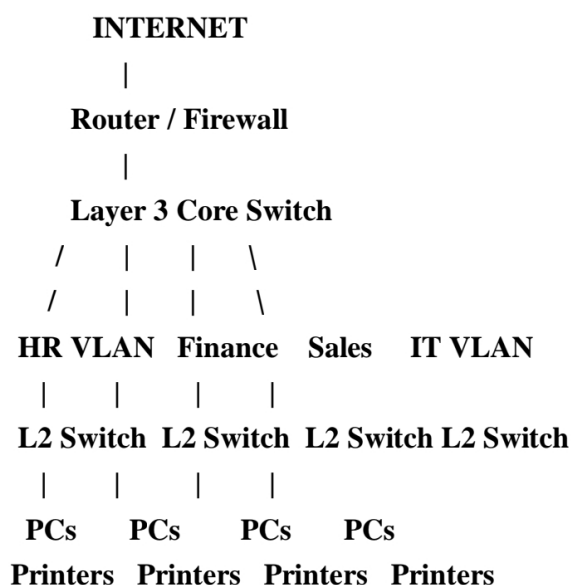
5. **Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing.**

Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

ANS:

Aim: To design a secure, scalable, and efficient network for a company with multiple departments such as HR, Finance, Sales, and IT.

Network Design:



Working

- Each department is assigned a separate VLAN.
- Layer 2 switches connect computers and printers within each department.
- The Layer 3 switch performs inter-VLAN routing.
- The router/firewall provides Internet access and security.
- VLANs reduce unnecessary broadcast traffic and provide logical separation.
- Access-control rules can restrict sensitive departments such as Finance.
- Redundant links can be added to improve network availability.

Advantages

1. Better network security.
2. Reduces broadcast traffic.
3. Easy department management.
4. Supports inter-department communication when permitted.
5. Easy to expand.
6. Efficient IP address management.
7. Centralized control and monitoring.

Example

If the Finance VLAN contains confidential financial systems, access-control rules can prevent ordinary Sales users from accessing those systems while still allowing authorized Finance and IT users to communicate with them.

Conclusion

A hierarchical VLAN-based network is suitable for companies with multiple departments because it provides security, performance, centralized management, scalability, and controlled communication between departments.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution